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Article *in* The Journal of Chemical Physics · December 2015

DOI: 10.1063/1.4938012

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AIP | The Journal of Chemical Physics

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Citation: The Journal of Chemical Physics **143**, 244707 (2015); doi: 10.1063/1.4938012
View online: http://dx.doi.org/10.1063/1.4938012
View Table of Contents: http://scitation.aip.org/content/aip/journal/jcp/143/24?ver=pdfcov
Published by the AIP Publishing

THE JOURNAL OF CHEMICAL PHYSICS **143**, 244707 (2015)

Interference between Coulombic and CT-mediated couplings in molecular aggregates: H- to J-aggregate transformation in perylene-based π-stacks

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(Received 9 October 2015; accepted 3 December 2015; published online 30 December 2015)

The spectroscopic differences between J and H-aggregates are traditionally attributed to the spatial dependence of the Coulombic coupling, as originally proposed by Kasha. However, in tightly packed molecular aggregates wave functions on neighboring molecules overlap, leading to an additional charge transfer (CT) mediated exciton coupling with a vastly different spatial dependence. The latter is governed by the nodal patterns of the molecular LUMOs and HOMOs from which the electron (t_e) and hole (t_h) transfer integrals derive. The sign of the CT-mediated coupling depends on the sign of the product $t_e t_h$ and is therefore highly sensitive to small (sub-Angstrom) transverse displacements or slips. Given that Coulombic and CT-mediated couplings exist simultaneously in tightly packed molecular systems, the interference between the two must be considered when defining J and H-aggregates. Generally, such π-stacked aggregates do not abide by the traditional classification scheme of Kasha: for example, even when the Coulomb coupling is strong the presence of a similarly strong but destructively interfering CT-mediated coupling results in "null-aggregates" which spectroscopically resemble uncoupled molecules. Based on a Frenkel/CT Holstein Hamiltonian that takes into account both sources of electronic coupling as well as intramolecular vibrations, vibronic spectral signatures are developed for integrated Frenkel/CT systems in both the perturbative and resonance regimes. In the perturbative regime, the sign of the lowest exciton band curvature, which rigorously defines J and H-aggregation, is directly tracked by the ratio of the first two vibronic peak intensities. Even in the resonance regime, the vibronic ratio remains a useful tool to evaluate the J or H nature of the system. The theory developed is applied to the reversible H to J-aggregate transformations recently observed in several perylene bisimide systems. © *2015 AIP Publishing LLC*. [http://dx.doi.org/10.1063/1.4938012]

I. INTRODUCTION

The nature of the electronic excitations in molecular aggregates has been a consistent theme in chemical physics since the discovery of J-aggregates by Scheibe[1] and Jelley[2] nearly a century ago. The UV-vis aggregate absorption spectrum is significantly modified by the presence of intermolecular interactions; generally, aggregates are classified as J-aggregates when the aggregate absorption peak is red-shifted from the monomer peak, and H-aggregates when it is blue-shifted. As proposed by Kasha,[3–6] the contrasting photophysical properties of J-aggregates and H-aggregates arise from the spatial dependence of the intermolecular Coulombic coupling. When the molecular transition dipole moments are parallel and "side-by-side," as exists in a sandwich type molecular dimer, the coupling is positive resulting in a H-aggregate. Conversely, when the transition dipole moments are oriented in a "head to tail" arrangement the coupling is negative, resulting in a J-aggregate. This basic assignment of J- and H-aggregates has prevailed in the community since its inception by Kasha over 5 decades ago. The most celebrated examples of J-aggregates (with presumably dominant Coulombic interactions) involve those composed of cyanine dyes such pseudoisocyanine chloride (PIC),[1,2] although certain perylene bisimides and porphyrins also form J-aggregates as detailed in a recent review.[7] More exotic nanotubular J-aggregates are also known.[8–10] On the other hand, H-aggregating chromophores include certain carotenoids,[11–13] oligothiophenes,[14,15] oligophenylene vinylenes,[14,16,17] and most perylene bisimides.[18–22] H-aggregation is also typical of polymer π-stacks such as those composed of poly(3-hexylthiophene).[23–26]

In many tightly packed molecular aggregates, such as π-stacks, nearest-neighbor separations are in the range of 3.3–4 Å. Molecules in such close proximity will interact via the usual Coulomb coupling, which generally drops off as a negative power of the intermolecular separation R, as well as a short-range coupling arising from intermolecular wave function overlap, which drops off exponentially with R. Coulombic interactions are responsible for Forster energy transfer,[27,28] while short-range interactions include Dexter and super-exchange transfer processes.[29–31] Historically, (singlet) excitons in molecular aggregates have been analyzed almost entirely in terms of long-range Coulombic coupling, which is often described using the point dipole approximation: for parallel transition dipole moments the Coulomb coupling is given by the expression,

$$J_{Coul} \approx \frac{\mu^2(1-3\cos^2\theta)}{4\pi\varepsilon R^3},$$

where θ is the angle between the transition dipole moment vector, $\boldsymbol{\mu}$, and the line connecting the molecular centers and ε is the medium dielectric constant. This expression is the source of the often-quoted axiom that "side-by-side" molecular orientations ($\theta = \pi/2$) lead to H-aggregates while "head-to-tail" orientations ($\theta = 0$) lead to J-aggregates. However, such an association fails when the short-range coupling is comparable or even larger than the Coulombic coupling. In such cases, the geometric realization of H- and J-aggregates depends on the spatial dependence of both the short-range and long-range couplings.

In a previous paper, we studied how HOMO-HOMO and LUMO-LUMO wave function overlap on neighboring chromophores gives rise to an efficient excitonic coupling mechanism which results in J- and H-aggregate-like photophysical behavior.[32] In the super-exchange mechanism, electron and hole transfer proceed through a virtual intermolecular CT state, effectively moving an excitation from one molecule to its neighbor.[30,31] When the CT state with energy E_{CT} is energetically well-separated from its parent Frenkel excitation with energy E_{S_1}, second order perturbation theory gives[32]

$$J_{CT} = \frac{-2D_e D_h}{E_{CT} - E_{S_1}}, \qquad \left|E_{CT} - E_{S_1}\right| \gg |D_e|\,,|D_h|$$

as the effective CT-mediated short-range coupling between nearest neighbors. J_{CT} depends on the product of the electron (D_e) and hole (D_h) dissociation integrals which are approximately equal to the electron (t_e) and hole (t_h) transfer integrals, respectively. In Ref. 32, we showed that charge transfer J-aggregates ($J_{CT} < 0$) and H-aggregates ($J_{CT} > 0$) display vibronic spectral signatures identical to the usual Coulomb-coupled J- and H-aggregates. However, the geometries which lead to charge-transfer J-aggregates and H-aggregates can be very different from those corresponding to Kasha J- and H-aggregates.

The dependence of J_{CT} on the product of $D_e D_h$ (or $t_e t_h$) makes J_{CT} highly sensitive to small intermolecular translations or "slips" within the π-stacks. Kazmaier and Hoffmann[33] and others[34–37] have shown that the conduction and valence band widths in perylene bisimide (PBI) π-stacks undergo strong deviations as one molecule is slipped past its neighbor. The bandwidth deviations, which occur on a sub-Å length scale, are associated with the nodal patterns of the overlapping HOMOs and LUMOs and are responsible for the remarkable range of pigment colors observed from crystal phases of perylene derivatives, a phenomenon known as crystallochromy.[33,35,38,39] A convincing demonstration of slip-sensitivity in a π-stacked PBI derivative was provided by Gregg and Kose,[40] who showed that the conversion from the thermodynamically stable black phase to the kinetically favored red phase is associated with just a 1.6 Å slip between neighboring molecules. The authors estimated that the sum of the valence and conduction bandwidths, $\approx 4(|t_e| + |t_h|)$, increases by ≈0.32 eV in going from the red-phase to the black-phase, quite a large change considering such a small intermolecular displacement.

In reality, short- and long-range couplings are simultaneously present in molecular π-stacks. As long as the diabatic CT and Frenkel bands remain energetically well-separated by an amount greater than $|t_e|$, and $|t_h|$, the total coupling between neighboring molecules is represented by the *sum*, $J_{Coul} + J_{CT}$.[32] In addition to further exploring how the interference between the two coupling sources impacts photophysical properties in the perturbative limit, another goal of the present paper is to study how the interference impacts these properties when the CT and Frenkel bands are closer to resonance. We have shown previously that the interference is destructive in 7,8,15,16 tetraazaterrylene (TAT) nanopillars[41] leading to a two-band line shape with properties of both J- and H-aggregates, due a to negative short-range CT-mediated couplings and positive long-range Coulomb couplings. Such "HJ" aggregates are in the intermediate coupling regime since $|t_e|$ and $|t_h|$ are comparable to $|E_{CT} - E_{S_1}|$. Although the second-order short-range coupling J_{CT} is not entirely justified in the intermediate coupling regime, it nevertheless provides a qualitative means of understanding the impact of Frenkel/CT mixing on photophysical properties. Indeed, the negative J_{CT} in TAT nearly cancels the positive nearest-neighbor Coulombic coupling, and the exciton bandwidth (calculated nonperturbatively) is dramatically reduced, leading to a substantial reduction in exciton mobility.[42]

The interference in the TAT "HJ"-aggregates described above is "integrated" because it derives from couplings between the *same* set of chromophores, i.e., between nearest neighbors in a π-stack. We distinguish this from our previous work on polymer π-stacks, where "HJ" aggregates resulted from opposing couplings between *different* sets of chromophores. For example, in polymer π-stacks the intrachain interactions are J-like while the interchain interactions are H-like.[43,44] In the molecular π-stacks considered in this work, the integrated interference can also be constructive (i.e., "JJ" or "HH") with profound effects on the absorption spectrum and transport properties.

In this paper we evaluate the impact of interfering short- and long-range couplings on the absorption spectrum of molecular π-stacks in the presence of vibronic coupling to the ubiquitous ring/vinyl stretching vibrational mode with energy ≈0.17-0.18 eV (1400 cm^{-1}) common to many conjugated organic molecules. The mode leads to pronounced vibronic progressions in the absorption spectrum of a single molecule in solution. Aggregation-induced changes to the regular progression are a source of additional signatures for H- and J-aggregation.[45] The starting point for our analysis is a Frenkel/CT Hamiltonian with local vibrational coupling. We analyze both the perturbative regime, where J_{CT} is of direct physical relevance, as well and the non perturbative regime where the diabatic CT and Frenkel bands are overlapping. We introduce the concept of a "null" aggregate in which the two coupling types effectively cancel, yielding a dispersionless exciton band; although the Coulomb coupling may be strong in null aggregates it is compensated by an equally strong but opposing short-range coupling, yielding an absorption spectrum closely resembling that of an unaggregated molecule in solution. Obviously, such aggregates strongly deviate from conventional Coulomb-coupled Kasha aggregates.

Applications are made to PBI π-stacks, where the coupling between Frenkel and CT excitons has been well-studied.[35,38,46–48] We are particularly interested in the so called H- to J-aggregate transformations in PBI complexes observed in the solution and solid states.[49,50] The reversible transformation, as identified through changes in the absorption spectrum, was induced in a core-unsubstituted PBI-melamine complex in Ref. 49 by the addition of cyanurate, and in a benzoylaminoethyl PBI complex in Ref. 50 by the addition of a linear alcohol. In both cases, the transformation was associated with a redistribution of intermolecular hydrogen bonding. The absorption spectrum switched between the "H-form," consisting of one main band, blue-shifted from the monomer with pronounced vibronic structure, to the "J-form," characterized by two broad bands, the lower of which is red-shifted from the monomer spectrum by several thousand wave numbers. In what follows we will demonstrate that the absorption spectrum of PBI π-stacks is due to a competition between short- and long-range couplings. The short-range component is extremely sensitive to translational displacements ("slips") and twists between neighboring chromophores while the long-range coupling is far less sensitive, allowing one to readily move through the null points which divide the J-and H-aggregate forms. We evaluate the absorption spectrum at geometries corresponding to local minima in the ground state energies determined through density functional theory (DFT) calculations by Fink *et al.*[51] Our analysis shows the J-form to be significantly displaced along the long molecular axis while the H-form is likely to undergo a significant angular twist.

II. HAMILTONIAN FOR π-STACKED AGGREGATES

In order to further our investigations of exciton photophysics in molecular π-stacks we employ a Hamiltonian which allows for both charge and energy transport in the presence of local vibronic coupling. To approach the essential physics in the most straightforward manner, we limit all couplings to nearest neighbors. The Hamiltonian represented in a subspace containing a single electron and a single hole can be divided as

$$H = H_{FE} + H_{CT}, \tag{1a}$$

where H_{FE} is the Frenkel/Holstein Hamiltonian,

$$\begin{aligned} H_{FE} = \hbar\omega_{0-0} + \sum_n J_{Coul}\{B_n^\dagger B_{n+1} + B_{n+1}^\dagger B_n\} \\ + \hbar\omega_{vib}\sum_n b_n^\dagger b_n + \hbar\omega_{vib}\sum_n \{\lambda_0(b_n^\dagger + b_n) + \lambda_0^2\}B_n^\dagger B_n, \end{aligned} \tag{1b}$$

sufficient to describe conventional Coulomb-coupled aggregates, and H_{CT} includes all terms arising from intermolecular charge separation,

$$\begin{aligned} H_{CT} = &\sum_n \{t_e c_n^\dagger c_{n+1} + t_h d_n^\dagger d_{n+1} + h.c.\} \\ &+ \sum_{n,r\neq 0} V_{CT}(r) d_n^\dagger d_n c_{n+r}^\dagger c_{n+r} \\ &+ \hbar\omega_{vib} \sum_{n,r\neq 0} \{\lambda_+(b_n^\dagger + b_n) + \lambda_-(b_{n+r}^\dagger + b_{n+r}) \\ &+ \lambda_-^2 + \lambda_+^2\} d_n^\dagger d_n c_{n+r}^\dagger c_{n+r}. \end{aligned} \tag{1c}$$

In H_{FE}, $B_n^\dagger(B_n)$ creates (annihilates) a local electronic (Frenkel) excitation—the state S_1—on the nth molecule with energy, $E_{S_1} = \hbar\omega_{0-0}$. The state S_1 is arrived at from the ground state S_0 by the creation of an electron in the LUMO and a hole in the HOMO, hence $B_n^\dagger \equiv c_n^\dagger d_n^\dagger$, where $c_n^\dagger(d_n^\dagger)$ creates an electron (hole) on the nth chromophore. H_{FE} includes the Coulombic coupling J_{Coul}, between nearest-neighbor chromophores as well as local vibronic coupling to the dominant intramolecular vibrational mode with energy, $\hbar\omega_{vib} \approx 1400$ cm^{-1}. The operators $b_n^\dagger$ and b_n, respectively, create and annihilate vibrational excitations on molecule n while the magnitude of the vibronic coupling is determined by the neutral exciton Huang-Rhys (HR) parameter, λ_0^2.

H_{CT} accounts for electron (t_e) and hole (t_h) transfer as well as the Coulombic binding between separated electrons and holes. The diabatic energy of the CT exciton consisting of an electron and hole separated by r ($\neq 0$) relative to a local Frenkel excitation is

$$\begin{aligned} V_{CT}(r) &= E_{CT} - E_{S_1} \qquad |r| = 1 \\ &= \infty \qquad\qquad\quad\; |r| > 1. \end{aligned} \tag{2}$$

E_{CT} is the energy of a bound CT pair, $E_{CT} \equiv I_p - E_A + E_P - e^2/4\pi\varepsilon_0\varepsilon_r d$, where I_p, E_A, and E_p represent the ionization potential, electron affinity, and polarization energy, respectively. The confinement of a separated electron/hole pair to adjacent sites simplifies the analysis of the Coulomb/CT-mediated coupling but does not significantly alter the basic photophysics.

CT states consisting of neighboring cations and anions are created by the dissociation of a locally excited Frenkel exciton. The Hamiltonian in Eq. (1c) accounts for this process through the transfer of an electron or hole away from the local excitation with coupling strength t_e or t_h, respectively. Hence, we are approximating the actual electron and hole dissociation integrals, D_e and D_h, with their one-electron approximations, t_e and t_h, respectively.[52] The quantity, $E_{CT} - E_{S_1}$, also influences the mixing between diabatic CT and Frenkel states, and is typically in the range of ±200 meV for PBI stacks.[35,38] Usually, $E_{CT} - E_{S_1}$ is adjusted to obtain good agreement between the calculated and measured absorption spectra.[35,38] First principle DFT calculations of CT state energies are also possible,[53–55] but require special care.[55] Finally, the last term in Eq. (1c) represents the linear cation- and anion-vibrational coupling, as modulated by the respective HR factors, λ_+^2 and λ_-^2.

Our model is similar to that employed by Yamagata *et al.* to study the important role played by CT states in the low-energy excitonic states in oligoacene crystals.[56] Similar approaches for treating both vibronic coupling and CT in perylene-based aggregates and crystals were developed by Hennessy and Soos,[57] Hoffmann *et al.*,[46,47] and Gisslen and Scholz.[35,38] In what follows, the Hamiltonian in Eq. (1) is expressed in a multi-particle basis set[58–61] consisting of Frenkel one- and two-particle states as well as two-particle nearest neighbor charge-separated states. Once expressed in this manner, the Hamiltonian is diagonalized numerically yielding all eigenstates and energies from which the absorption spectrum is constructed according to Eq. (S.1) (see supplementary material for complete details[77]). Unless otherwise noted, the homogenous line width for all subsequent calculations was taken as $\Gamma = 0.28\hbar\omega_{vib}$.

For a linear array of N chromophores with periodic boundary conditions, all excitonic eigenfunctions of H in Eq. (1) can be described with a wave vector k, which takes on the values $k = 0, \pm 2\pi/N, \pm 4\pi/N, \ldots, \pi$. k is in units $1/d$ where d is the nearest neighbor separation. In the important limit of no vibronic coupling ($\lambda_0^2 = 0$), the diabatic CT excitons form two N-fold degenerate bands having energy E_{CT}. The CT states take the form[62]

$$|CT,k,\pm\rangle = \frac{1}{\sqrt{2N}} \sum_n e^{ikn} \{|n,n+1\rangle \pm |n,n-1\rangle\}, \quad (3)$$

where $|n,n+1\rangle$ indicates the charge-separated state with a cation (or hole) on molecule n and an anion (or electron) on molecule $n+1$, while all other sites are in the electronic ground state (i.e., in S_0). The Frenkel states are similarly delocalized; a Frenkel exciton with wave vector k is given by

$$|k\rangle = \frac{1}{\sqrt{N}} \sum_n e^{ikn} |n\rangle . \quad (4)$$

Here, $|n\rangle$ indicates that the nth chromophore is excited to the state S_1, while all other chromophores remain unexcited. The energies of the diabatic Frenkel states in (4) obey the dispersion relation

$$E_F(k) = E_{S_1} + 2J_{Coul}\cos(k). \quad (5)$$

Once t_e and t_h are activated, CT and Frenkel excitons with the same wave vector mix, leading to three eigenstates for each wavevector k: one of the resulting eigenstates remains a pure (antisymmetric) CT state leading to a dispersionless CT band (with energy E_{CT}), while the other two eigenstates are of mixed Frenkel/CT character as determined by t_e, t_h, E_{CT}, and J_{Coul}. For example, when $k = 0$, the eigenstates are composed of the symmetric CT state, $|CT, k = 0, +\rangle$, and the $|k = 0\rangle$ Frenkel state. For a given k, the Frenkel/CT mixing results in the dispersion relation

$$E_\pm(k) = E_F(k) + \frac{E_{CT} - E_F(k)}{2} \pm \sqrt{\left(\frac{E_{CT} - E_F(k)}{2}\right)^2 + 2(t_e^2 + t_h^2 + 2t_e t_h \cos k)} \quad (6)$$

for the upper ($E_+(k)$) and lower ($E_-(k)$) bands. In what follows, we concern ourselves only with the Frenkel/CT bands, ignoring the dispersionless CT band. The latter does not contribute to the absorption spectrum, as we further assume that the diabatic CT states carry no oscillator strength.

Throughout the remainder of the paper, we utilize the free-exciton dispersion bands in Eq. (6) as a guideline for understanding dispersion and spectral properties in the presence of vibronic coupling. One of the most important features of Eq. (6) is the profound dependence of $E_\pm(k)$ on the relative sign between t_e and t_h. For example, the two $k = 0$ states responsible for absorption undergo far greater splitting when t_e and t_h have the same sign as opposed to the case when they have opposite signs.[35,38] This is most dramatic when the magnitudes of t_e and t_h are similar; when $t_e = -t_h$ the ($k = 0$) diabatic Frenkel and CT excitons do not mix at all. As shown in Ref. 32, the relative sign also controls the H and J-like properties induced by CT, as will be further explored below.

Eq. (6) also predicts the existence of dispersionless bands. Such "flat" bands develop when the Coulombic coupling satisfies the "null" relation, $J_{Coul} = J_{Coul}^\pm$ where

$$J_{Coul}^\pm = \frac{4t_e t_h}{(E_{CT} - E_{S_1}) \pm \sqrt{(E_{CT} - E_{S_1})^2 + 8(t_e^2 + t_h^2)}}. \quad (7)$$

Inserting either $J_{Coul} = J_{Coul}^+$ or $J_{Coul} = J_{Coul}^-$ into Eq. (6) yields a flat band,

$$E_1(k) = E_{CT} - 4t_e t_h / 2J_{Coul}^\pm, \quad (8a)$$

as well as a band which retains the dispersion of the Frenkel exciton,

$$E_2(k) = E_F(k) + 4t_e t_h / 2J_{Coul}^\pm. \quad (8b)$$

For other values of the Coulomb coupling, curvature exists in both bands. In Secs. III–VI, we explore how the absorption spectrum changes as one passes through a null point. Generally, passage though the null point is associated with a H- to J-aggregate transformation (or vice versa).

III. WEAK FRENKEL/CT EXCITON COUPLING

We first examine the limit of weak Frenkel/CT exciton coupling—the limit in which the dissociation integrals, Coulomb couplings and vibrational mode energies are all small compared to the energetic separation between the diabatic CT and local Frenkel exciton states

$$|t_e|, |t_h|, |J_{Coul}|, \hbar\omega_{vib} \ll |E_{CT} - E_{S_1}|. \quad (9)$$

This limit is important theoretically as it affords a dramatic simplification of the Hamiltonian to an effective Frenkel-Holstein Hamiltonian as detailed in Ref. 32. This limit also offers novel aggregate photophysics, including the existence of a null aggregate.

When condition (9) is satisfied, the energetically distant CT state acts as a virtual intermediate in a "superexchange" process whereby an exciton is transferred from one chromophore to its neighbor. The effective Frenkel exciton coupling is given by[32]

$$J_{CT} = \frac{-2t_e t_h}{E_{CT} - E_{S_1}} \quad (10)$$

so that the total Frenkel exciton coupling between neighboring chromophores is $J_{Coul} + J_{CT}$. Hence, the absorption spectrum responds to an interference between the Coulombic and CT-mediated couplings, leading to interesting photophysical behaviors. This remarkable result calls into question the traditional H- and J-aggregate assignment based on geometric arguments arising from the Coulomb term alone (i.e., side-by-side is H and head-to-tail is J), as J_{CT} and J_{Coul} have very different spatial dependences (see Section V). The interference between couplings additionally offers the possibility of a null point signifying a H- to J-transition, whenever

$$J_{Coul} = -J_{CT} = 2t_e t_h / (E_{CT} - E_{S_1}) \quad (11)$$

so that the total coupling between nearest neighbors vanishes. Eq. (10) is also obtained as the limiting behavior of the null condition in Eq. (7) when $|E_{CT} - E_{S_1}| \gg |t_e|, |t_h|$.

The impact of the interference between J_{Coul} and J_{CT} is demonstrated in Fig. 1, which shows a series of calculated spectra in which the Coulomb coupling increases from a negative value, through zero, and then to positive values in going from top to bottom. In all spectra, we took equal electron and hole dissociation integrals, $t_e = t_h = 1$ and a CT energy given by $E_{CT} - E_{S_1} = 10$. (All energies are expressed in units of vibrational quanta $\hbar\omega_{vib} = 1400$ cm^{-1}.) The inset of each panel in Fig. 1 contains the lowest-energy exciton band. The $k = 0$ exciton gives rise to the first vibronic peak in the absorption spectrum. In the figure, the HR factor for an individual chromophore is set to unity, so that the first two vibronic peaks (A_1, A_2) have equal intensity when intermolecular coupling is absent ($t_e = t_h = J_{Coul} = 0$), as is the case for the single molecule spectrum shown as a dashed line in each of the panels (a)-(e). For a single molecule the vibronic peak intensities A_1, $A_2, \ldots$ scale as $A_{n+1} = \exp(-\lambda^2)\lambda^{2n}/n!$ (where $n = 0, 1, 2 \ldots$). Fig. 1 also shows an amplified region of the absorption spectrum surrounding the energy of the diabatic CT state, E_{CT}. States residing in this region are dominated by the CT component; the very small Frenkel admixture renders such states weakly absorbing, as we have assumed that the diabatic CT states carry no oscillator strength. For the remainder of this section, we focus mainly on the lower energy states dominated by the Frenkel component.

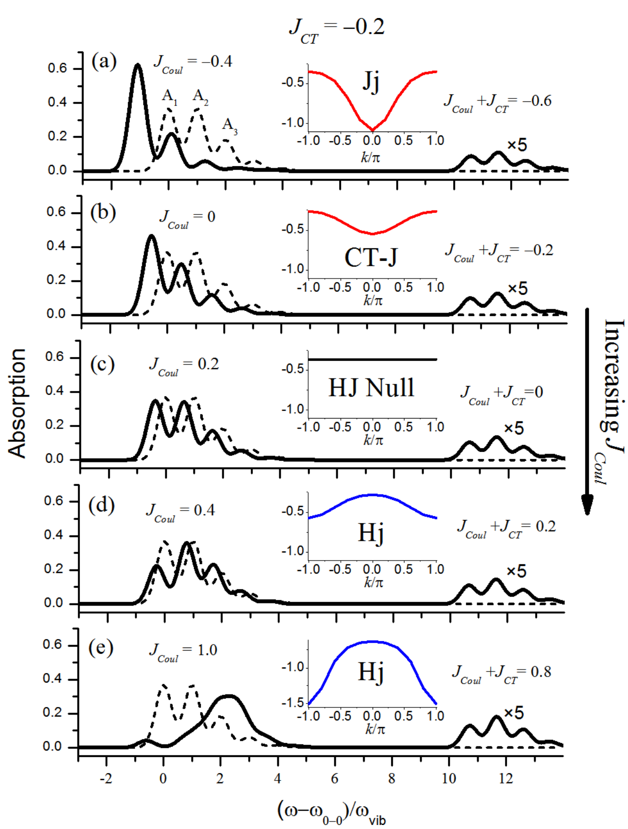

FIG. 1. Calculated absorption spectra (black) for π-stacks containing $N = 10$ chromophores based on the Hamiltonian in Eq. (1a) with CT parameters, $t_e = t_h = 1$ and $E_{CT} - E_{S1} = 10$ in units of vibrational quanta, $\hbar\omega_{vib} = 1400$ cm^{-1}. The CT-mediated coupling, $J_{CT} = -0.2$, is evaluated from Eq. (10). In panels (a)-(e), the Coulomb coupling increases from −0.4 to 1.0 as indicated. Vibronic coupling is described with an HR factor, $\lambda_0^2 = 1$ so that the A_1 and A_2 peaks are of equal intensity in the monomer spectrum (dashed). The ionic HR factors are each 0.5. Insets show the band dispersion for the lowest energy exciton.

Based on the parameters of Fig. 1 and Eq. (10), the CT mediated interactions should induce J-like behavior, as both $t_e t_h$ and $E_{CT} - E_{S_1}$ are greater than zero. Indeed, Eq. (10) gives $J_{CT} = -0.2$. In Fig. 1(a), the Coulombic coupling is also negative ($J_{Coul} = -0.4$) leading to a "reinforced" J-aggregate. The absorption spectrum is red-shifted with a significantly enhanced A_1/A_2 peak intensity ratio compared to the monomer spectrum, as expected for J-aggregates.[45] The spectral properties are consistent with the form of the exciton band which has positive curvature, placing the $k = 0$ exciton at the band bottom. The bandwidth is due to the sum of both coupling sources ($J_{CT} + J_{Coul}$) and is therefore enhanced by roughly 50% compared to a Kasha aggregate with only Coulomb coupling. We refer to such an aggregate as a Jj-aggregate, where the first letter indicates that J_{Coul} is negative and the second letter indicates that J_{CT} is negative. The lower case "j" further indicates that $|J_{CT}|$ is substantially less than $|J_{Coul}|$. In the Kasha limit $|J_{Coul}| \gg |J_{CT}|$, one can neglect the second letter, giving the conventional "J-aggregate" notation.

Fig. 1(b) shows that even when J_{Coul} is zero the spectrum retains its J-like qualities due entirely to Frenkel/CT mixing, i.e., though the negative J_{CT}. In such a CT J-aggregate, the exciton band curvature remains positive, although smaller than in Fig. 1(a) due to the absence of Coulomb coupling. A CT J-aggregate is spectrally indistinguishable from a Kasha J-aggregate, but due to the nature of the coupling, may not abide by the Kasha geometries expected for H- and J-aggregates, as detailed in Section V.

The interference between the two coupling types is most profound in Fig. 1(c), in what we refer to as a "Null" aggregate. Here, the (positive) Coulomb coupling exactly cancels the (negative) CT-mediated coupling, $J_{Coul} + J_{CT} = 0$, causing the exciton band to completely flatten. The absorption spectrum of such an HJ null aggregate strongly resembles the monomer spectrum, but for a small, though not insignificant, red-shift which is addressed below. Subsequent increases in J_{Coul} beyond the null point result in an increasingly negative band curvature. A negative curvature defines a H-aggregate, where the $k = 0$ state resides at the top of the band. The spectrum eventually blue-shifts relative to the monomer and the A_1/A_2 intensity ratio drops below the monomer value, as is characteristic of conventional (Coulomb-coupled) H-aggregates.[45] Note that in Figs. 1(d) and 1(e) the two coupling sources interfere destructively, leading to a bandwidth that is compromised relative to that expected of a traditional Kasha aggregate where only J_{Coul} is present.

In Fig. 2, we show a similar series of spectra but with one very important difference: t_e and t_h are now taken to be out of phase, i.e., $t_e = -t_h = 1$. According to Eq. (10), J_{CT} is now positive, with $J_{CT} = +0.2$. To best appreciate the interference between the two coupling types, we assign a positive value for J_{Coul} in the top panel and continually decrease it through zero in going from top to bottom. Hence, in Fig. 2(a) the Coulomb and CT-mediated coupling are both positive leading to a reinforced Hh-aggregate. The spectral characteristics are the same as those put forth by Kasha: a strong, blue shifted main absorption peak and a negative band curvature. The

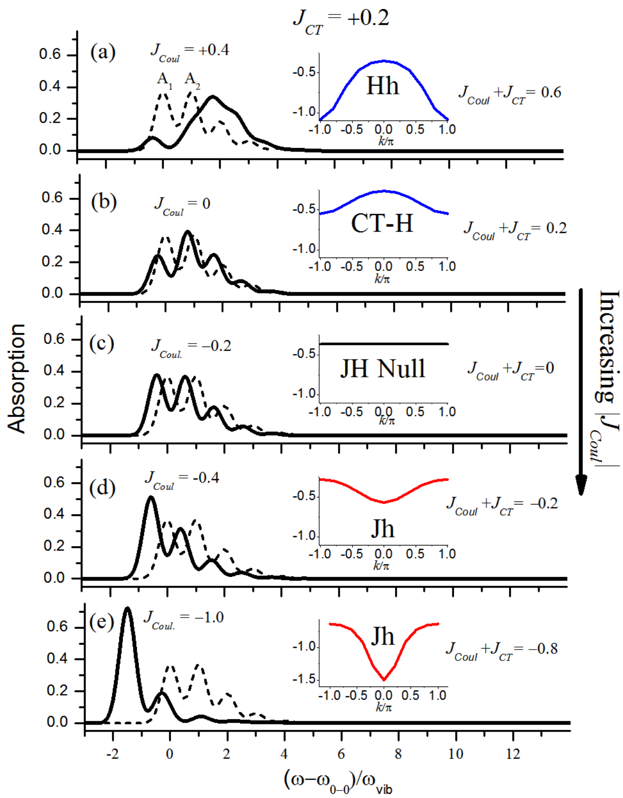

FIG. 2. Calculated absorption spectra (black) for π-stacks containing $N = 10$ chromophores based on the Hamiltonian in Eq. (1a) with CT parameters, $t_e = -t_h = 1$ and $E_{CT} - E_{S_1} = 10$ in units of vibrational quanta, $\hbar\omega_{vib} = 1400$ cm^{-1}. The effective coupling, $J_{CT} = +0.2$, is evaluated from Eq. (10). In panels (a)-(e), the Coulomb coupling decreases from 0.4 to −1.0 as indicated. Vibronic coupling is described with an HR factor, $\lambda_0^2 = 1$ so that the A_1 and A_2 peaks are of equal intensity in the monomer spectrum (dashed). The ionic HR factors are each 0.5. Inset shows the band dispersion for the lowest energy exciton.

vibronic signatures also reflect H-aggregation with a greatly attenuated A_1/A_2 ratio. Decreasing the Coulomb coupling to zero, as in Fig. 2(b), results in a weakly coupled CT H-aggregate, due entirely to the positive CT-mediated coupling, J_{CT}. Interestingly, although the band curvature is negative and the A_1/A_2 intensity ratio is less than the monomer value (as defines H-aggregates), *the spectrum remains red-shifted relative to the monomer*, as opposed to the blue-shift present in Kasha H-aggregates. This very peculiar difference between CT H-aggregates and Kasha H-aggregates is due to the way the band curvature is established in CT-mediated aggregates as discussed below.

Fig. 2(c) shows that, once again, a null point is encountered when the two coupling types exactly cancel. As in Fig. 1(c), the spectrum is essentially identical to the monomer spectrum but for the small red-shift. Subsequent deceases in J_{Coul} eventually lead to the strong-red-shifted Kasha-like J-aggregate spectrum in Fig. 2(e), hosting a large positive band curvature and a greatly enhanced A_1/A_2 ratio over the monomer value. Note, however, that due to the destructive nature of the coupling interference, the exciton bandwidths in Figs. 2(d) and 2(e) are compromised relative to a true Kasha J-aggregate with only Coulomb coupling.

All of the behaviors described above can be readily accounted for by an effective Frenkel-Holstein Hamiltonian. In the perturbative regime defined by condition (9), the effective Hamiltonian takes the form

$$\begin{aligned} H_{eff} &= (\hbar\omega_{0-0} + \Delta_{CT}) \sum_n B_n^\dagger B_n \\ &+ \sum_n (J_{Coul} + J_{CT})(B_n^\dagger B_{n+1} + B_{n+1}^\dagger B_n) \\ &+ \hbar\omega_{vib} \sum_n b_n^\dagger b_n + \hbar\omega_{vib} \sum_n \{\lambda_0 (b_n^\dagger + b_n) + \lambda_0^2\} B_n^\dagger B_n, \end{aligned} \tag{12}$$

where the J_{CT} is given in Eq. (10) and Δ_{CT} is a self-energy correction given by

$$\Delta_{CT} = -2 \frac{(t_e^2 + t_h^2)}{E_{CT} - E_{S_1}}. \tag{13}$$

Δ_{CT} was derived in Ref. 32 for a dimer, which lacks the factor of two in Eq. (13) due to both chromophores having only a single nearest neighbor. Although not shown, we have verified that the effective Hamiltonian of Eq. (12) essentially reproduces the spectra in Figs. 1 and 2.

The spectral shift Δ_{CT} in Eq. (13) accounts for all second order interactions in which the virtual CT state is formed from electron (hole) transfer upon which the initial local Frenkel excitation is reformed by a second electron (hole) back-transfer. Δ_{CT} is universally negative whenever the diabatic CT state is higher in energy than the local Frenkel exciton, as exists in both Figs. 1 and 2, and it successfully accounts for the red–shifts observed in the null aggregates therein (panels (c)). Due to the form of Δ_{CT} in Eq. (13), the red shift persists for *both* CT J-aggregates and CT H-aggregates, as observed in Figs. 1(b) and 2(b) when the Coulomb coupling is zero. Hence, in CT H-aggregates a red-shift occurs despite a negative band curvature, quite different from the conventional Kasha mechanism operative in Coulomb-coupled H-aggregates. Interestingly, if $E_{CT} - E_{S_1}$ is instead negative, then Δ_{CT} becomes positive, promoting a *blue shift* for both CT J- and H-aggregates.

The unusual photophysical behaviors displayed in Figs. 1 and 2 can be better understood by appealing to the energy level diagrams in Figs. 3 and 4. For simplicity, vibronic coupling is ignored, so that the band energies reported in the figure insets correspond to Eq. (6). In Fig. 3, the formation of a hJ-aggregate from a Coulomb-defined H-aggregate is depicted for the simple case when t_e and t_h are equal, so that the CT-mediated coupling is J-like. The diabatic CT band is N-fold degenerate with energy E_{CT}, while the diabatic Frenkel band portrays a positive value for the Coulombic coupling—i.e., a conventional Coulomb-coupled H-aggregate with negative band curvature. The width of the red arrows tracks the magnitude of the coupling between CT and Frenkel states with the same wave vector k, equal to $|t(1 + e^{-ik})|$ when $t = t_e = t_h$. Thus, the coupling is maximal between states with $k = 0$ and is equal to zero for states with $k = \pi$. When the charge transfer coupling, t, is turned on, the energy levels assume the ordering depicted in Fig. 3(b). Since, in this example, $|J_{CT}|$ surpasses J_{Coul}, the ordering of the states in the lower band is completely reversed upon Frenkel/CT mixing, in effect converting a Kasha H-aggregate into a CT J-aggregate. As shown earlier, all of the spectral signatures are

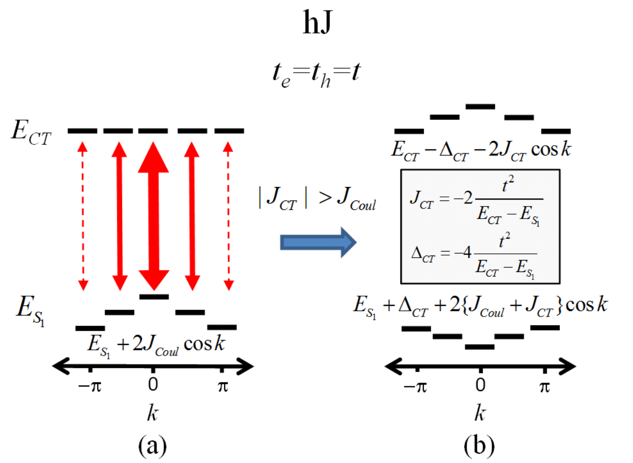

FIG. 3. (a) Energy level diagram depicting the interaction of diabatic CT and Frenkel exciton bands in an hJ aggregate. The interaction strength is proportional to the arrow widths. Here, $J_{Coul} > 0$, so that the diabatic Frenkel band is H-like (negative curvature). (b) Exciton bands after Frenkel/CT mixing with $t_e = t_h$. The in-phase relationship between t_e and t_h leads to $J_{CT} < 0$. The figure depicts the case, $|J_{CT}| > J_{Coul}$, so that the lower band is overall J-like (positive curvature). Note: If $|J_{CT}| = J_{Coul}$ the lower band in (b) would be flat.

consistent with this scenario. Note also that if the CT-mediated coupling exactly compensated the Coulomb coupling, i.e., if $J_{CT} + J_{Coul} = 0$, the resulting adiabatic band would be entirely flat as in a Null aggregate.

Fig. 3(b) also shows the upper band surrounding E_{CT}, the $k = 0$ state of which is responsible for the weak, high energy absorption features found in Fig. 1 due to its minority Frenkel component. The negative band curvature, consistent with H-aggregation, is also consistent with the vibronic line shape of the absorption band in the high-energy region of Fig. 1. Careful consideration shows that the upper band is dictated by an effective coupling, $-J_{CT}$, which in Fig. 3 is positive. In the weak coupling regime of the present section, the upper band shape is independent of J_{Coul}. Such properties are evident in Fig. 1.

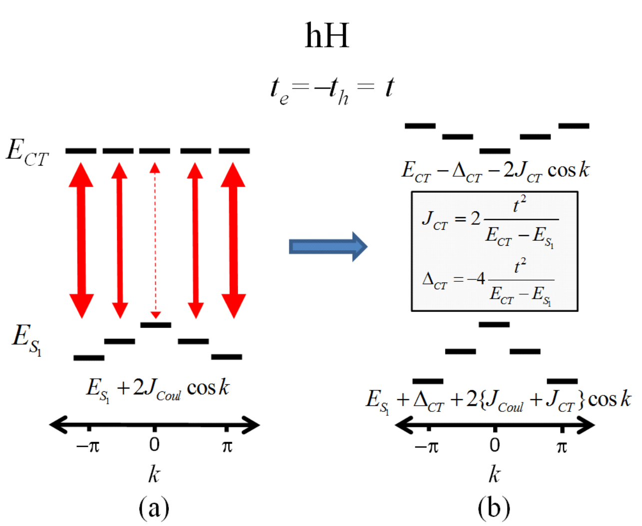

FIG. 4. (a) Energy level diagram depicting the interaction of diabatic CT and Frenkel exciton bands in an hH aggregate. The interaction strength is proportional to the arrow widths. Here, $J_{Coul} > 0$, so that the diabatic Frenkel band is H-like (negative curvature). (b) Exciton bands after Frenkel/CT mixing with $t_e = -t_h$. The out of phase relationship leads to $J_{CT} > 0$. Since both couplings are positive (H-like) the resulting lower exciton band in (b) remains H-like with an enhanced bandwidth compared with (a).

Finally, let us consider the effect of changing the sign of the product $t_e t_h$ on the energy level diagram in Fig. 3. In Fig. 4, we set $t = t_e = -t_h$ so that J_{CT} becomes positive, indicating that CT-mediated interactions will reinforce H-aggregate behavior. In this case, the magnitude of the k-dependent coupling between diabatic CT and Frenkel excitons is $|t(1 - e^{-ik})|$. In a dramatic departure from the situation in Fig. 3, the maximal coupling is now between states with $k = \pi$, with the coupling between the $k = 0$ states identically zero. Hence, the negative band curvature is enhanced making the aggregate even more H-like but for one spectral property: As depicted in the figure, both the diabatic and adiabatic $k = 0$ states have the same energy. This arises because the blue shift driven by the positive value of J_{CT} is exactly compensated by the red-shift due to Δ_{CT}. Hence, the CT-mediated couplings do not cause a spectral blue-shift of the $k = 0$ exciton, as occurs in the Coulomb-coupled aggregates. Rather, they induce a *red-shift* for excitons which increases with the value of $|k|$. Such a mechanism also accounts for the odd red-shift presented by the CT H-aggregate in Fig. 2(b). In this case the red-shift Δ_{CT} is not completely compensated by the blue shift in J_{CT} as it is in Fig. 4 (where vibronic coupling is ignored). The effective Hamiltonian in Eq. (12) shows that the red-shift Δ_{CT} is unencumbered by Franck Condon factors, which is not the case for J_{CT}, resulting in a red-shift for the CT-mediated H-aggregate.

IV. BEYOND PERTURBATION THEORY: ONE-BAND VS. TWO-BAND AGGREGATES

As the CT energy approaches the diabatic Frenkel exciton band, the superexchange mechanism based on the second-order expressions in Eqs. (10) and (13) and visualized in Figs. 1-4 begins to break down. Results based on the effective Frenkel-Holstein Hamiltonian in Eq. (12) become more approximate as the CT and Frenkel states begin to strongly mix. This is demonstrated in Fig. 5 for the cases $t_e = t_h = 1$ (a)-(e) and $t_e = -t_h = 1$ (f)-(j). In both cases, the CT band was chosen to be only two vibrational quanta above molecular excitation energy, i.e., $E_{CT} - E_{S_1} = 2$, only twice the value of $|t_e|$ or $|t_h|$. The HR factor and vibrational energy remain identical to the value used in Figs. 1 and 2. The figure shows that the relative phase between the two dissociation integrals continues to a have a dramatic impact on the resulting spectra. When $t_e = t_h$ (Figs. 5(a)-5(e)) there are two prominent absorption bands, in marked contrast to the single main band resulting when t_e and t_h have opposite signs (f)-(j). The single or double band nature can be understood by examining the matrix elements coupling the (diabatic) $k = 0$ vibronic Frenkel excitons, which carry the bulk of the oscillator strength, and the (diabatic) $k = 0$ CT states, which have no oscillator strength.[35] This is most easily appreciated when vibronic coupling is absent. In this case, the matrix element coupling the "free" excitons, $|k = 0\rangle$ and $|CT, k = 0, +\rangle$ in Eqs. (3) and (4) is equal to $\sqrt{2}(t_e + t_h)$, so that the band splitting is $2\sqrt{2}|t_e + t_h|$. When the magnitude

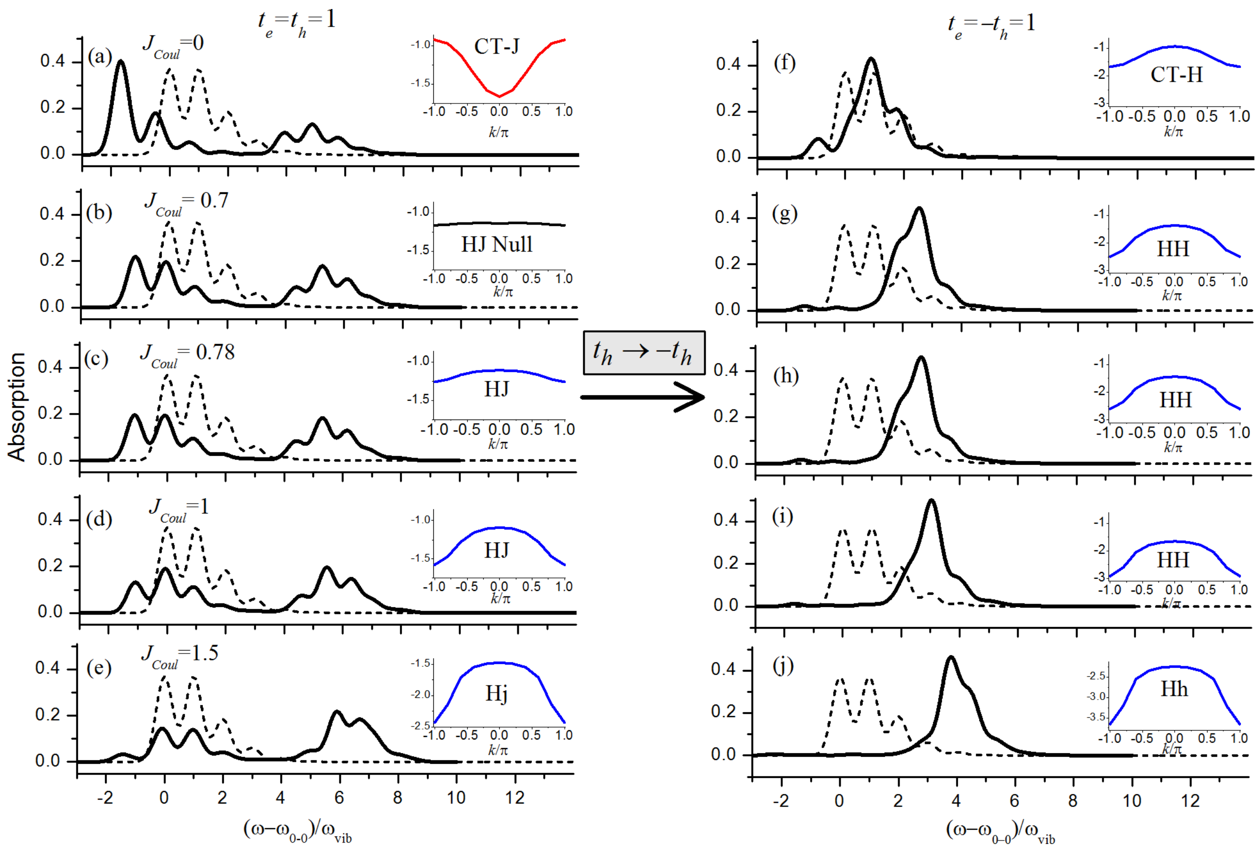

FIG. 5. Calculated absorption spectra (black) for π-stacks containing $N = 10$ chromophores based on the Hamiltonian in Eq. (1a) with, $t_e = t_h = 1$ (a)–(e) and $t_e = -t_h = 1$ (f)-(j). In all cases, $E_{CT} - E_{S_1} = 2$. In panels (a)-(e) (and (f)-(j)) the Coulomb coupling increases from 0 to 1.5 with the values indicated. Vibronic coupling is described with an HR factor, $\lambda_0^2 = 1$ so that the A_1 and A_2 peaks are of equal intensity in the monomer spectrum (dashed). The ionic HR factors are each 0.5. Insets show the band dispersion for the lowest energy exciton.

of the matrix element is large compared to $|E_{CT} - E_{S_1}|$, the oscillator strength is effectively distributed over both bands. Conversely, when the matrix element is small compared to $|E_{CT} - E_{S_1}|$, as is the case when the CT integrals are out of phase (Figs. 5(g)-5(j)), there is little mixing amongst the $k = 0$ states and oscillator strength is confined to a single band. This is also evident in the perturbative regime of Fig. 2, where the high-energy region surrounding the diabatic CT band is devoid of any discernable absorption. The influence of Frenkel/CT exciton coupling on the overall absorption line width was studied in detail by Gisslen and Scholz[35,38] for a series of perylene based dye aggregates, where larger bandwidths were attributed to enhanced values of $|t_e + t_h|$.

Despite being beyond the perturbative regime, there are still several features in the spectra of Fig. 5 that can be understood, at least qualitatively, in terms of interfering CT-mediated and Coulombic couplings. For example, in the spectra ((a)-(e)), the CT mediated coupling given by Eq. (10) is $J_{CT} = -1$ and, in the absence of Coulomb coupling (Fig. 5(a)), the low-energy band is clearly J-like. As in Sec. III, increasing the Coulomb coupling leads to a null point at which the band becomes dispersionless (Fig. 5(b)). However, the null point occurs when $J_{Coul} \approx 0.7$ and not 1 as predicted in the perturbative limit. The value of J_{Coul} at the null point can be better understood by considering the limit of no vibronic coupling. According to Eq. (7), the condition at which the lower band becomes flat (when $E_{CT} - E_{S_1} > 0$) is given by

$$J_{Coul} = \frac{4t^2}{(E_{CT} - E_{S_1}) + \sqrt{(E_{CT} - E_{S_1})^2 + 16t^2}} \qquad t_e = t_h = t. \tag{14}$$

When $E_{CT} - E_{S_1} \gg |t|$, Eq. (13) reduces to the perturbation theory result, where the null point for the lower band occurs when $J_{Coul} = 2t^2/(E_{CT} - E_{S_1})$. When the parameters from Fig. 6 ($E_{CT} - E_{S_1} = 2$ and $t = 1$) are inserted into Eq. (14) the null point becomes $J_{Coul} = 0.62$, in much better agreement with the value of 0.7 from the figure. The remaining small discrepancy is due to vibronic coupling.

Fig. 5(b) also shows that at the null point the A_1/A_2 ratio is not exactly equal to the monomer value as it is in the perturbation regime of Fig. 1(b). In Fig. 5(b), the ratio is about 10% larger. The ratio becomes the monomer value (which is unity in the current example) only when the Coulomb coupling is slightly increased to 0.78 as shown in Fig. 5(c), at which point the band curvature becomes slightly negative. Further increases in J_{Coul} result in an increase in the bandwidth, and the low energy band becomes H-like. The spectrum responds accordingly, with the A_1/A_2 ratio driven significantly below the monomer value as in Fig. 5(d). Interestingly, the upper band, which is dominated by CT excitons, has negative curvature throughout (not shown) making the high-energy

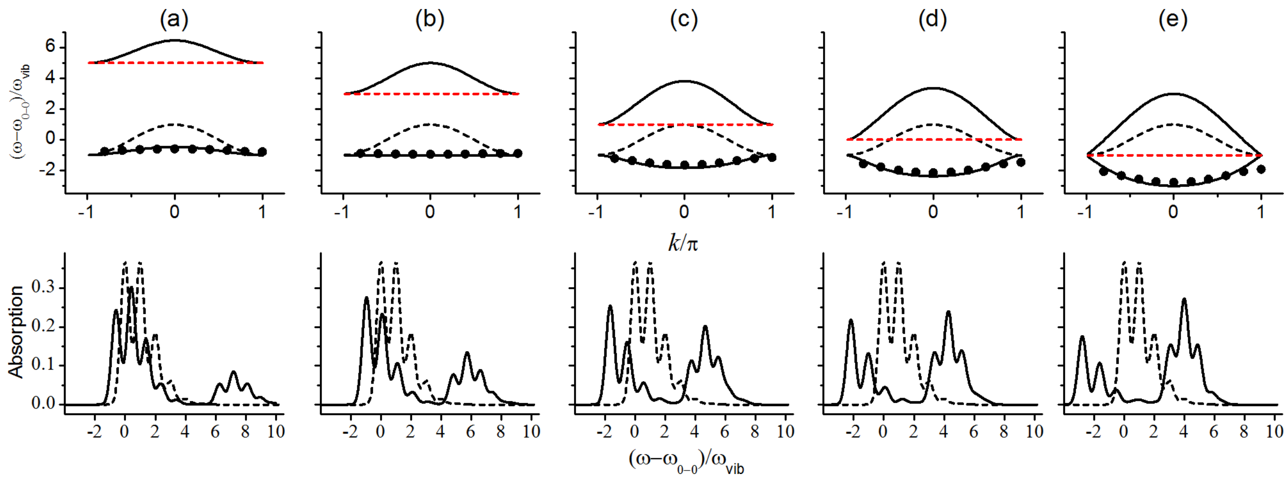

FIG. 6. Calculated exciton band dispersions (upper) and associated absorption spectra (lower) for π-stacks containing $N = 10$ chromophores. Calculated spectra are based on the Hamiltonian in Eq. (1a) using $t_e = t_h = 1$, $J_{Coul} = 0.5$, $\lambda^2 = 1$ and with ionic HR factors of 0.5. In moving from left to right, the diabatic CT energy (red dash) is lowered through the diabatic Frenkel band (black dash) resulting in the Frenkel/CT bands (black solid) evaluated using the free-exciton dispersion in Eq. (6). The black circles represent the calculated lower band dispersion in the presence of vibronic coupling.

band in the absorption spectrum H-like in all of the spectra ((a)-(e)), very similar to what happens in the perturbation limit.

When the sign of the hole dissociation integral is negated ($t_e = -t_h$), the series of spectra in Figs. 5(f)-5(j) result. The spectra are now dominated by a single band due to the lack of Frenkel/CT mixing between the $k = 0$ states ($|CT, k = 0, +\rangle$ and $|k = 0\rangle$). The curvature of the lower energy band is universally negative as would also be predicted in the perturbative limit since $J_{CT} = +1$, thereby reinforcing the positive Coulomb couplings throughout the series and leading to enhanced H-like behavior. Indeed, all spectra in Figs. 5(f)-5(j) have a reduced A_1/A_2 intensity ratio and a strongly blue-shifted main absorption peak.

In order to better appreciate the approach to resonance, we show in the top panel of Fig. 6 the upper and lower free-exciton energy bands as E_{CT} is varied while all other parameters are held constant. Here, we consider the case of in-phase electron and hole transfer integrals ($t_e = t_h = 1$) and hold the Coulomb coupling at the value $J_{Coul} = 0.5$. The dashed curves represent the diabatic Frenkel (black) and CT (red) bands, while the black solid curves are the resulting lower and upper exciton bands after admixture. The free-exciton energy bands omit vibronic coupling (all HR factors are zero) so that they are solutions to Eq. (6). To demonstrate the effect of vibronic coupling, we also show the lower energy band (black circles) when the HR factors are nonzero ($\lambda^2 = 1;\ \lambda_+^2 = \lambda_-^2 = 0.5$). In this example, the diabatic Frenkel band is assumed H-like (negative curvature) but a similar figure can be composed for Coulombic J-aggregates in a straightforward manner.

Since t_e and t_h are in phase ($t_e = t_h$), the $k = 0$ states interact most strongly, as in Fig. 3, so that as the diabatic CT band approaches from above (Fig. 6(a)), the adiabatic lower energy band begins to flatten as the curvature becomes less negative. The null point is reached in Fig. 6(b) when $E_{CT} - E_{S_1}$ is approximately 3. This value is essentially equal to that computed without vibronic coupling, as can be seen by rearranging the null condition in Eq. (7) to read

$$E_{CT} - E_{S_1} = \frac{2t^2 - 2(J_{Coul})^2}{J_{Coul}} \qquad t_e = t_h = t \tag{15}$$

and inserting the parameters from Fig. 6 ($t = 1$ and $J_{Coul} = 0.5$), giving $E_{CT} - E_{S_1} = 3$. The flat band in Fig. 6(b) agrees with Eq. (8a) using $J_{Coul} = 0.5$. As can be further appreciated from Figs. 6(a), 6(c)-6(e) outside the null region, vibronic coupling generally reduces the bandwidth by increasing the exciton's effective mass. As the diabatic CT band is lowered beyond the null point, the adiabatic lower band develops an increasingly larger *positive* curvature (c)-(e), in effect, having undergone a H to J conversion. Conversely, the adiabatic upper band always develops a negative curvature, which continues to increase in magnitude throughout the series (a)-(e).

The lower panels of Fig. 6 show the associated two-band absorption spectra including vibronic coupling. The band-to-band separation is dominated by Frenkel/CT splitting, which is close to the free-exciton value of $2\sqrt{2}(t_e + t_h) = 5.66$ and remains fairly stable throughout the series. As expected, the spectral area of the lower absorption band is initially (Figs. 6(a)-6(c)) greater than that of the second band due to the larger $k = 0$ Frenkel character in the mixed state. However, the situation reverses once the diabatic CT band passes through resonance with the $k = 0$ Frenkel exciton (Figs. 6(d) and 6(e)) as the upper band retains the majority of the $k = 0$ Frenkel exciton character. Interestingly, when the diabatic CT band is degenerate with the $k = 0$ Frenkel exciton (Fig. 6(c)), the mixed states have equal Frenkel/CT character so that the spectral area is divided evenly between the upper and lower absorption bands. The vibronic ratios involving the first two peaks in each band are fairly good indicators of the respective band curvatures. For example, in Fig. 6(a), the ratio of the first two peaks in each band is attenuated relative to the monomer value (which is unity) reflecting the negative curvature of each band (H-like). In Fig. 6(e), the lower band is J-like while the upper is H-like and the vibronic ratios respond accordingly: the lower band hosts a ratio greater than the monomer value (J-like) while the upper band hosts a ratio less than the monomer value (H-like). However, the ratio is not exactly the monomer value at the null point (Fig. 6(b)), as was the case in Fig. 5.

Because J_{CT} can still be defined as long as $E_{CT} - E_{S_1}$ is not zero, we can continue to use the two-letter aggregate notation developed above. Hence, the aggregates in Figs. 6(a)-6(d) can be described as HJ aggregates, while that in Fig. 6(e) is a HH aggregate due to the sign change in $E_{CT} - E_{S_1}$.

V. APPLICATIONS TO PBI π-STACKS: H- TO J-AGGREGATE TRANSFORMATIONS

We now apply the theory developed above to the H- to J-aggregate transformations observed in several PBI derivatives in solution and in the solid state.[49,50] In Ref. 49, the transformation was induced in a (core-unsubstituted) PBI-melamine complex through a change in the hydrogen bonding, mediated by the addition of a cyanuric acid derivative. The UV-Vis absorption spectra corresponding to the J- and H-aggregate forms, as well as the unaggregated monomer (M) in solution, are reproduced in Fig. 7(a). The monomer spectrum (black dash) consists of a well-defined Franck-Condon progression, which can be fit with an HR factor of approximately 0.7. The spectrum shown in red is associated with the J-aggregate form because of the emergence of a low-energy band, red-shifted by approximately 3000 cm^{-1} from the main monomer peak. However, there is also a broader, blue-shifted band, peaked at roughly 22 000 cm^{-1}. Both bands are symmetrically positioned around the main monomer peak. In marked contrast, the spectrum in blue hosts a single band with vibronic structure, the peak of which is modestly blue-shifted by approximately 800 cm^{-1} from the monomer peak. Hence, the blue spectrum is assumed to arise from a H-aggregate. Consistent with this assignment is the substantially diminished A_1/A_2 peak ratio compared to the monomer. A similar transformation for a H-bonding PBI derivative was recently observed by Sarbu *et al.*[50] in the solid phase, where the addition of an organic non solvent triggered the H to J transition.

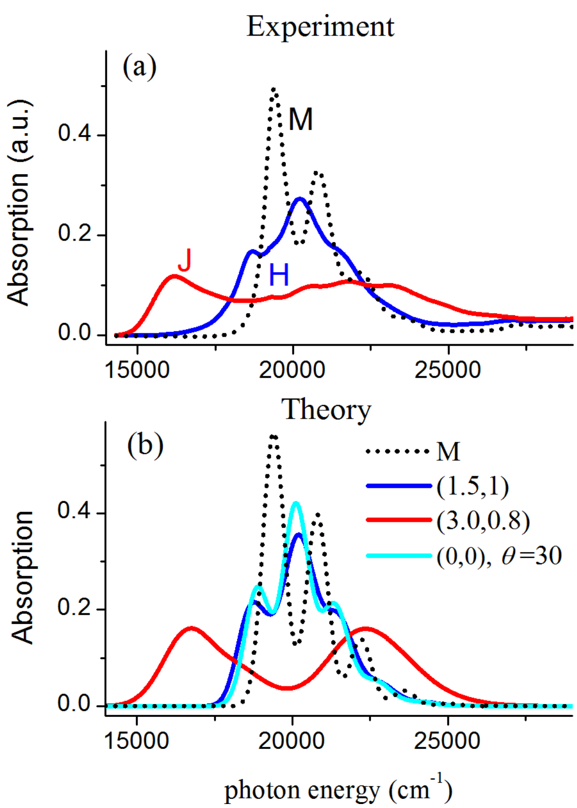

FIG. 7. (a) Measured absorption spectra for the monomer in solution (M) and the J- and H-aggregated forms of the PBI-melamine complexes from Ref. 49. (b) Simulated monomer absorption spectra (black dash) using an HR factor of 0.7 and vibrational frequency, $\hbar\omega_{vib} = 1400$ cm^{-1}. The three solid spectra correspond to π-stacks containing $N = 10$ perylene chromophores based on the Hamiltonian in Eq. (1a) using $E_{CT} - E_{S_1} = 0$, and t_e and t_h obtained from Fig. 8 for the nearest-neighbor orientations indicated in the figure inset. The two slip stacks correspond to the local ground state minima calculated in Ref. 51, while the twisted stack is the global minimum, also determined from Ref. 51. The Coulomb coupling is taken from Fig. 8 after screening with a dielectric constant of 3.0. The ionic HR factors are each 0.35. The Gaussian homogeneous line width is $\Gamma = 0.28$ (blue and cyan spectra) and $\Gamma = 0.56$ (red spectrum). Finally, all calculated aggregate spectra were red-shifted by 500 cm^{-1} (solution-to-crystal shift).

As described in Ref. 49, neighboring chromophores in the J-stack are believed to be significantly slipped relative to each other along the long molecular axis, by perhaps 3-4 Å. The reasoning is based on Kasha theory and the spatial properties of the Coulomb coupling. Accordingly, in the H-aggregate the slip is believed to be much smaller. In both aggregate forms, a relative twist between neighboring chromophores has also been suggested; indeed helical π-stacks based on chiral substituted PBI's exhibit absorption spectra very similar to the H-form in Fig. 7(a).[22] Detailed DFT calculations by Fink *et al.*[51] have shown that in closely packed PBI dimers there exist two local ground-state minima which correspond to slip-translations at approximately (3.5,1) and (1.5,1), where the first and second numbers indicate, respectively, the long- and short-axis displacements in Angstroms relative to an eclipsed configuration. Subsequent DFT calculations by Vura-Weis *et al.*[34] support the existence of two stable regions at approximately the same locations as Fink *et al.*[51] for a host of PBI derivatives. According to Ref. 51 the *global* minimum corresponds to a configuration with no relative translation at all, (0,0), but rather a relative twist of approximately 30° around an axis connecting the mass-centers.

In order to show how the competition between short- and long range interactions in perylene based π-stacks impact the absorption spectrum in such a dramatic manner as observed in Refs. 49 and 50, we begin with Fig. 8 which shows a contour plot of the calculated values (DFT/B3LYP, cc-pVDZ) for t_e and t_h as a function of slip distance along the short and long molecular axes for two unsubstituted perylenes separated by 3.5 Å. The electron and hole transfer integrals are obtained from

$$t_e \equiv \langle LUMO_A | H | LUMO_B \rangle, \tag{16a}$$

$$t_h \equiv -\langle HOMO_A | H | LUMO_B \rangle, \tag{16b}$$

where H designates the one-electron terms in the electronic dimer Hamiltonian which includes chromophores A and B. The values of t_e and t_h are in good agreement with those calculated in Refs. 33 and 35. Due to the HOMO and LUMO nodal patterns in perylene, t_h undergoes three sign changes upon longitudinal displacements $(x,0)$ while t_e does not change sign at all.[33] Displacements along the short-axis,

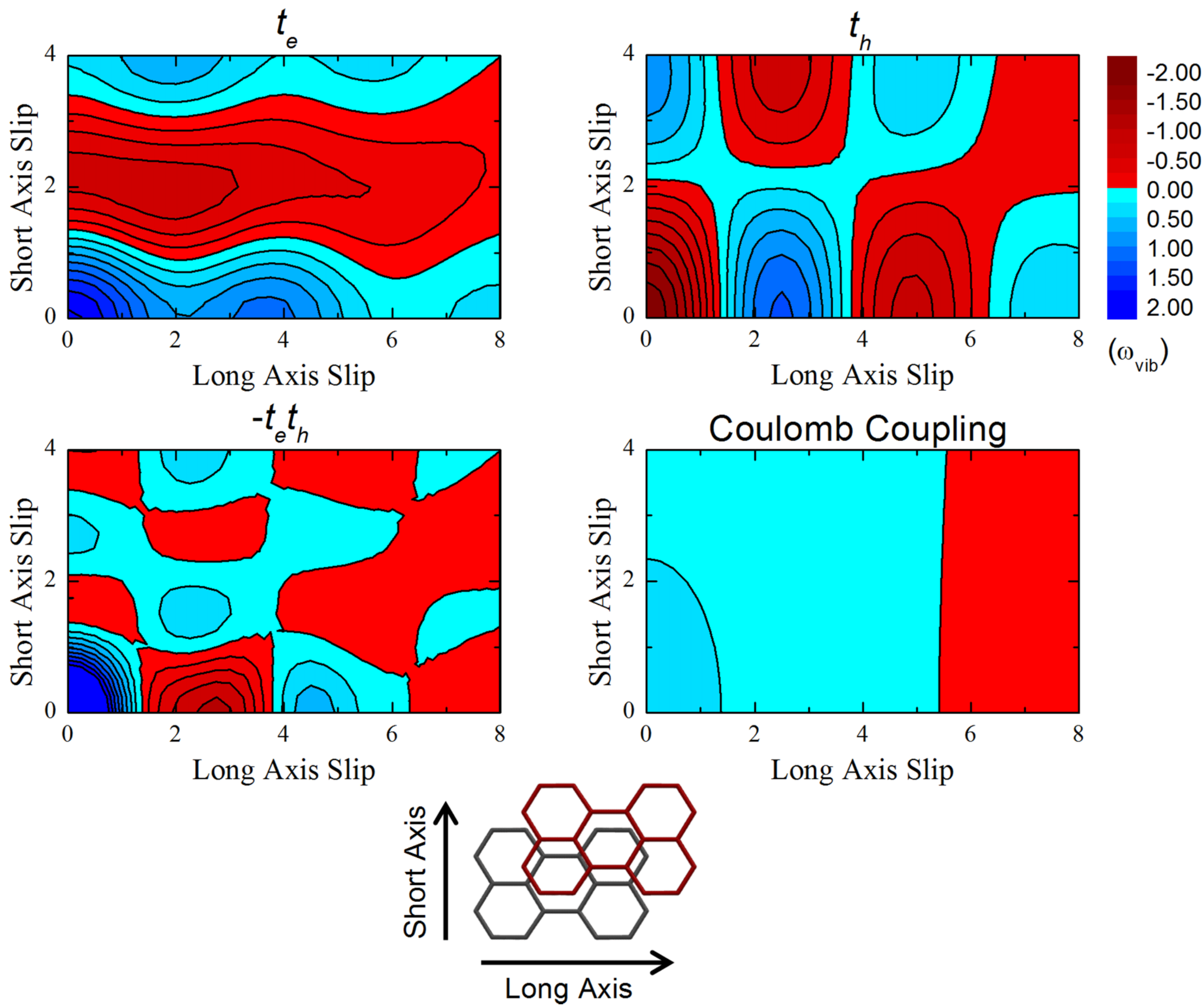

FIG. 8. (Top) Calculated electron and hole transfer integrals between two perylene molecules as a function of transverse displacement. The intermolecular separation is 3.5 Å. (Bottom) The product $-t_e t_h$ and the Coulomb coupling as a function of transverse displacement (see text for details).

$(0, y)$, induce a sign change in t_h and three sign changes in t_e (a third sign change which occurs around 5 Å is not shown in Fig. 8(a)). The peculiar nodal patterns which exist in t_e and t_h are fully consistent with the findings of Kazmaier and Hoffmann.[33]

In Fig. 8(c) we show how the product $-t_e t_h$ behaves as a function of slip translation, in order to track the short-range coupling, J_{CT}. In the red colored regions $-t_e t_h$ is less than zero, so that J_{CT} in Eq. (10) is negative and therefore J-promoting when $E_{CT} - E_{S_1} > 0$. Conversely, in the blue-colored regions the sign for $-t_e t_h$ is positive, indicating H-like behavior. (For negative values of $E_{CT} - E_{S_1}$ the H and J assignment is reversed.) Finally, Fig. 8(d) shows the unscreened nearest-neighbor Coulomb coupling, as calculated using atomic transition charge densities (TD-DFT/B3LYP, cc-pVDZ).[63–65]

Fig. 8 shows that the spatial scales for the interconversions between H- and J-like behaviors driven by the couplings, J_{CT} and J_{Coul} are quite different. In the Kasha picture, which is based on Coulomb coupling alone (Fig. 8(d)), a change in the sign of the coupling requires a displacement along the long molecular axis of over 5 Å. This is consistent with the early work of Czikklely *et al.*[66] where it was shown that a longitudinal displacement slightly over half the length of the molecule is required to change the sign of the Coulomb coupling for two parallel rod-shaped molecules separated by distances smaller than their lengths (i.e., where the point-dipole approximation breaks down).[67] In marked contrast, the CT-mediated coupling, J_{CT}, (Fig. 8(c)) changes sign three times in each direction, with the first occurring for a displacement of only 1.3 Å along either the long axis or the short axis. Hence, it is possible for aggregates to display spectral and electronic characteristics that directly contrast what is expected of traditional Kasha aggregates at a given geometry.

To better appreciate the markedly different spatial dependencies presented by J_{CT} and J_{Coul}, we show in Fig. 9 the dependence of the various couplings along a longitudinal slip $(x, 0)$. J_{CT} was evaluated from Eq. (10) using $E_{CT} - E_{S_1} = 5$. The figure also plots the total coupling $J_{CT} + J_{Coul}/\varepsilon$ which determines J or H-aggregate behavior in the perturbative limit. Here, the unscreened Coulomb coupling J_{Coul} is reduced by the spatially averaged optical dielectric constant of PBI, which

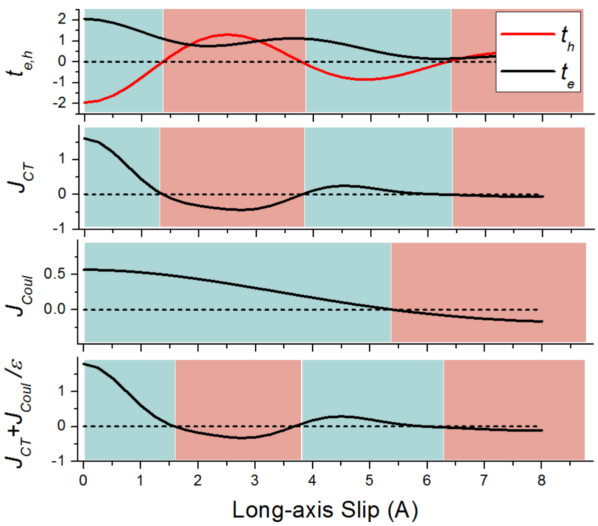

FIG. 9. Calculated electron and hole transfer integrals and electronic couplings between two perylene molecules as a function of longitudinal displacement, assuming an intermolecular separation is 3.5 Å. To evaluate J_{CT}, Eq. (9) was used with $E_{CT}-E_{S_1}=5$. The total coupling in the bottom panel was evaluated assuming a dielectric constant, $\varepsilon=3$.

we take to be $\varepsilon = 3$.[68] Interestingly, the CT-mediated coupling dominates over most of the slip range, so that the total coupling changes sign (interconverts between J and H-aggregate behavior) at roughly the same points that J_{CT} changes sign.

Fig. 10 shows the calculated perylene absorption spectrum at select points along the long-axis slip for two cases. In the bottom panel, $E_{CT}-E_{S_1}=5$, so that the perturbative limit holds. By comparing to the monomer spectrum (dashed curves) it is easy to appreciate H-aggregate behavior in the blue regions and J-aggregate behavior in the red regions. It is very important to note that H-like (J-like) behavior is indicated by a A_1/A_2 ratio that is diminished (enhanced) *relative to the monomer value*. Since we have used an HR factor of 0.7, the A_1/A_2 ratio in the monomer is 1.43.

For many perylene-based dyes, it is more likely that the CT and Frenkel states are closer to resonance, with the resulting Frenkel/CT mixing promoting excimer like formation consistent with the emissive behavior of many PBI aggregates.[69–71] In order to account for this important nonperturbative regime we also evaluate the absorption spectra for $E_{CT}-E_{S_1}=0$, shown in the top panel of Fig. 10. In the resonant regime, the much stronger Frenkel/CT mixing is expected to result in a two-band or one-band spectrum, depending on the relative magnitudes of t_e+t_h and t_e-t_h. This is indeed the case; when the slip distance is only 0.5 Å, the antisymmetric sum dominates, $|t_e-t_h| \gg |t_e+t_h|$, and the spectrum consists of a single main peak, blue shifted from the monomer spectrum. For slip distances between approximately 2 and 3 Å $|t_e+t_h|$ dominates, yielding two distinct bands with a separation between the band maxima approximately given by the free-exciton value, $2\sqrt{2}|t_e+t_h|$. Interestingly, the large red shift of the lower energy band, which has led researchers to label such species as "J-aggregates" has nothing to do with the Coulombic coupling, as in Kasha theory. Rather, it is due to the strong coupling between Frenkel and CT excitons. In the vicinity of a 5 Å slip, $|t_e-t_h|$ once again dominates yielding a single absorption band. Finally, after yet another reversal beyond approximately 6 Å the two-band spectrum is recovered, but with a significantly diminished splitting compared to the spectrum at 3 Å. This type of merged two-band structure resembles what is found in TAT nanopillars.[41]

We next consider the electron and hole dissociation integrals as well as the Coulombic coupling as a function

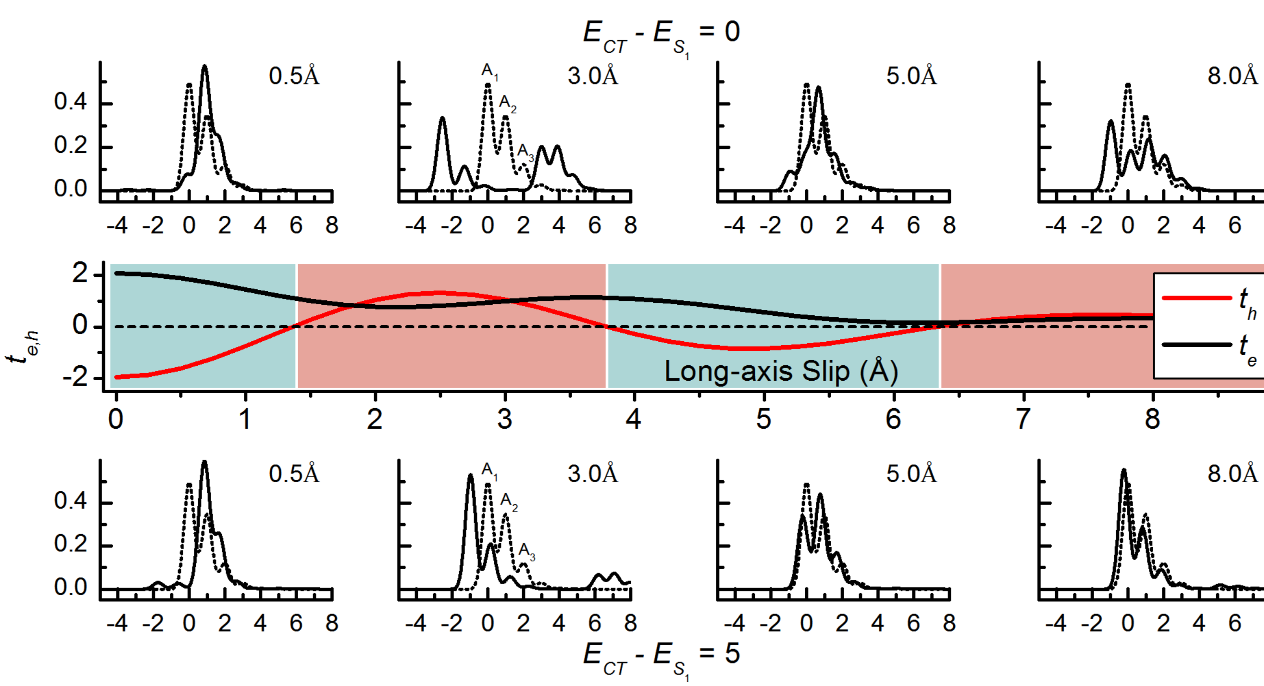

FIG. 10. Calculated absorption spectra at various longitudinal slips for a π-stack containing N = 10 perylene molecules using the full Hamiltonian in Eq. (1a). In the top series of spectra, the diabatic CT energy is resonant with the molecular excitation energy, $E_{CT}-E_{S_1}=0$, while in the bottom series $E_{CT}-E_{S_1}=5$. The HR factor was set at 0.7 so that the A_1 peak dominates the monomer spectrum (dashed). Electronic couplings in the aggregate spectra were taken from Fig. 8. Ionic HR factors were 0.35 each.

of the twist angle, θ, for a perylene dimer with interfacial separation of 3.5 Å. The results, shown in Fig. 11, are in good agreement with Ide *et al.*[37] Interestingly, for twist angles close to 30°, t_e and t_h are *both* zero, completely inactivating charge transfer. At this value of θ, Frenkel excitons and CT excitons cannot mix and the chromophores interact entirely by Coulombic coupling, which, according to Fig. 11(b), leads to a conventional Kasha H-aggregate. Fig. 12 shows the absorption spectra evaluated at selected angles. Indeed, for $\theta = 30°$ the spectrum resembles that of a Coulomb-coupled H-aggregate, as was also shown by Fink *et al.*[51] Based on the signs of t_e and t_h in Fig. 11 a transformation to a two-band spectrum is expected beyond approximately 48°. This is verified in Fig. 12 which shows a pronounced two-band spectrum at 60°. Hence, one can interconvert between J- and H-aggregate behavior by changing the twist angle.

We conclude this section by comparing the spectra calculated for the three geometries which minimize the total ground state energy as determined by Fink *et al.*[51] with the J- and H- aggregate spectra of Yagai *et al.*,[49] Fig. 7(b) shows the results. The geometry with the greatest longitudinal displacement (3,0.8) is associated with large, in-phase values of the electron and hole transfer integrals (t_e and t_h are, respectively, 606 and 1248 cm^{-1}) resulting in a pronounced two-band spectrum, which agrees quite well with the J-aggregate form of Ref. 49 (red curve in Fig. 7(a)). In particular, the peak to peak separation is quantitatively reproduced. Upon transforming to the other two geometries, the spectrum dramatically changes to an H-aggregate form, which can be reconciled quite well using Kasha theory, as done in Ref. 51. At the point (1.5, 1) the electron and hole transfer integrals are quite weak, 347 and −20 cm^{-1} respectively,

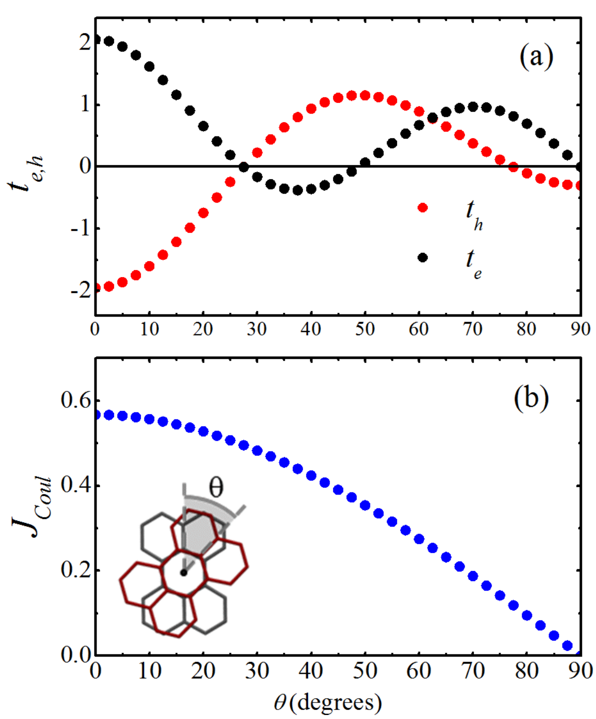

FIG. 11. (a) The electron and hole transfer integrals as a function of twist angle between two unslipped perylene molecules separated by 3.5 Å. (b) The unscreened Coulomb coupling as a function of twist angle.

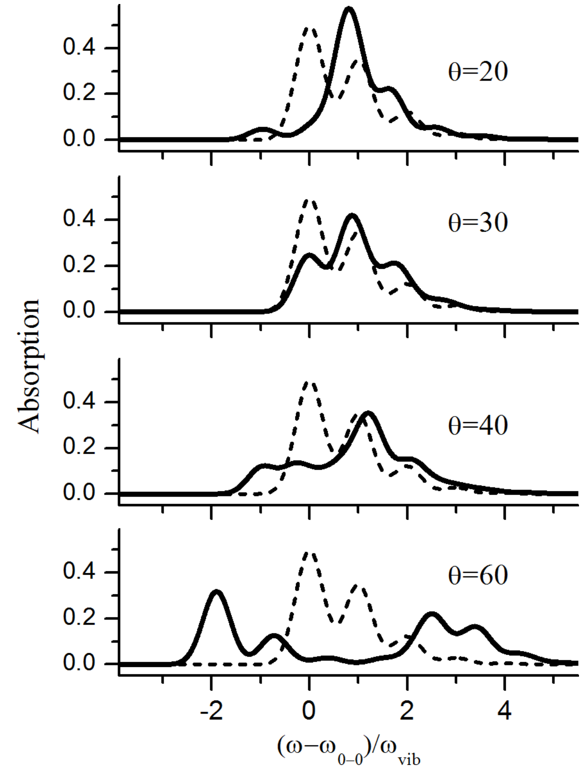

FIG. 12. Calculated absorption spectrum for several values of the twist angle, θ, in a perylene π-stack containing 10 molecules. The Hamiltonian in Eq. (1a) was used with $E_{CT} - E_{S_1} = 0$ and t_e, t_h and J_{Coul} (screened by $\varepsilon = 3$) taken from Fig. 11. The HR factor is $\lambda_0^2 = 0.7$, while the ionic HR factors are 0.35 each. Dashed spectrum is the monomer ($N = 1$) spectrum.

allowing the positive Coulomb coupling to dominate. As mentioned earlier, at the twist angle of 30° only the positive Coulomb coupling exists. Both calculated spectra agree very well with the observed H-aggregate PBI spectrum shown in Fig. 7(a).

VI. DISCUSSION/CONCLUSION

Our primary purpose was to demonstrate interference effects between CT-mediated and Coulombic couplings in molecular π-stacks and their relationship to the J- to H-aggregate transformations observed in several PBI derivatives.[49,50] In order to arrive at the essential photophysics in the most direct manner we utilized a pared-down electronic Hamiltonian—one that includes only nearest-neighbor Coulombic coupling (a rather good approximation in one-dimension) and limits electron-hole separation to neighboring chromophores. Including local vibronic coupling involving the ubiquitous 1400 cm^{-1} ring stretching mode allowed us to track H- and J-aggregate behavior through vibronic spectral signatures. As outlined in Refs. 45 and 32 the ratio of oscillator strengths between the first two vibronic absorption peaks increases (decreases) relative to the monomer ratio in Coulomb coupled and CT mediated J- (H-) aggregates. These results are generalized in the present manuscript for integrated Coulomb/CT aggregates. We find that the vibronic ratio rigorously tracks the band curvature in the perturbative limit: a peak ratio greater than the monomer ratio indicates a positive band curvature (band minimum at $k = 0$) as *defines* J-aggregates, while a peak ratio less than the monomer ratio indicates a negative band curvature (band maximum at $k = 0$) as *defines* H-aggregates.[72] Even in the resonance regime, we find the vibronic ratio strongly correlates to the band curvature.

The interference between the CT-mediated (superexchange) and Coulombic couplings is most easily appreciated in the perturbative limit where optical absorption is accurately described using an effective Holstein-style Hamiltonian (see Eq. (12)). In this limit, the total coupling between nearest neighbors is simply $J_{Coul} + J_{CT}$, with J_{CT} given by Eq. (10). Hence, the ratio formula, originally derived for Coulomb-coupled Frenkel aggregates,[25,26,45] can be generalized by replacing the Coulomb coupling with the total coupling

$$\frac{I_{A_1}}{I_{A_2}} \approx \frac{1}{\lambda_0^2}\frac{[1 - 2e^{-\lambda^2}G(0,\lambda_0^2)(J_{Coul} + J_{CT})/\hbar\omega_{vib}]^2}{[1 - 2e^{-\lambda^2}G(1,\lambda_0^2)(J_{Coul} + J_{CT})/\hbar\omega_{vib}]^2}, \quad (17)$$

where I_{A_1} and I_{A_2} are the line strengths for the first and second vibronic peaks and

$$G(v,\lambda_0^2) \equiv \sum_{\substack{u=0,1,2,..\\(u\neq v)}} \lambda_0^{2u}/[u!(u-v)].$$

In the limit of $\lambda_0^2 = 1$, (16) reduces to

$$\frac{I_{A_1}}{I_{A_2}} \approx \frac{[1 - 0.96(J_{Coul} + J_{CT})/\hbar\omega_{vib}]^2}{[1 + 0.29(J_{Coul} + J_{CT})/\hbar\omega_{vib}]^2}. \quad (18)$$

Hence, for a J-aggregate, ($J_{Coul} + J_{CT} < 0$), the ratio becomes greater than the monomer value while in a H-aggregate ($J_{Coul} + J_{CT} > 0$), the ratio is less than the monomer value.

The effective Hamiltonian leads naturally to the existence of so-called "null-aggregates" where the total coupling is equal to zero, i.e., total destructive interference between the Coulomb and CT-mediated couplings. In a null aggregate, the exciton band is flat and Eq. (17) predicts that the aggregate will exhibit a ratio exactly matching the monomer value. In fact, in the perturbation regime, the entire absorption spectral line shape for a null aggregate is essentially identical to the monomer spectrum but for a small red shift due to a second order correction to the band energy, Δ_{CT} in Eq. (13). In null aggregates, the exciton mobility is also severely compromised.[42]

Interestingly, even when the diabatic CT and Frenkel bands are close to resonance the concept of a null aggregate persists. This is readily appreciated from the free-exciton dispersion in Eq. (6), which predicts flat bands in any coupling regime when the condition in Eq. (7) is satisfied. For example, when the diabatic CT energy is resonant with the center of the diabatic Frenkel band, ($E_{CT} - E_{S_1} = 0$) a flat band occurs when

$$J_{Coul} = \pm\frac{\sqrt{2}t_e t_h}{\sqrt{t_e^2 + t_h^2}}, \quad (19)$$

and the flatness persists even when vibronic coupling is activated (see Fig. 6).

In the perturbative regime, passage through the null point, effected by changing the sign of either the electron or hole transfer integrals, t_e or t_h (so that J_{CT} changes sign) is accompanied by a J/H-aggregate transformation as evidenced primarily by a change in I_{A_1}/I_{A_2} compared to the monomer value. In the resonant or near-resonant regime, the change is far more dramatic with the absorption spectral line shape evolving from a two-band shape (the "J" form) to a single band ("H" form), or vice versa. The change is most pronounced when the $k = 0$ free exciton splitting, $2\sqrt{2}|t_e + t_h|$, is large compared to a vibrational quantum so that both bands of the "J" form are well resolved. Single-band line shapes result when the $k = 0$ free exciton splitting becomes small compared to a vibrational quantum, which can arise if t_e and t_h remain large in magnitude but carry opposite signs ("H" form), or simply if $|t_e|$ and $|t_h|$ both become small compared a vibrational quantum so that the two bands of the "J" form are indistinguishable. The latter scenario is most likely relevant to the single band spectra observed in the "H" aggregate forms of Refs. 49 and 50. In these systems, the single absorption band is H-like because the (positive) Coulombic coupling, typical of side-by-side packing, dominates the CT-mediated coupling in either the slipped configuration at (1.5,1) or the twisted configuration ($\theta = 30$). Assuming the slipped configuration, a H to J transformation occurs with an increase in the longitudinal slip by only 1.5 Angstroms (see Fig. 7(b)). The scenario so described is consistent with the work of Gregg and Kose[40] who showed a reversible transformation from a "black" phase to a "red" phase in an oriented film of liquid crystal PBI molecules upon exposure to water vapor. The absorption spectrum of the red-phase strongly resembles the H-form in Refs. 49 and 50 while the black phase resembles the J-form. Interestingly, XRD measurements showed the red-to-black phase (H to J) transition is associated with a 1.6 Å transverse slip, consistent with our estimate. (Unfortunately, the direction of the slip in the transverse plane could not be determined in Ref. 40.) Moreover, Gregg and Kose concluded that the red phase aggregate (H-form) is due primarily to Frenkel excitons while the red-phase (J-form) arises from mixed Frenkel/CT excitons, in full agreement with our analysis.

It is likely that the J- to H-transformation may also be accompanied by a relative twist between the perylene chromophores. Analysis of electron diffraction data by, Sarbu *et al.* supported a H-aggregate form consisting of helical π-stacks with a relative twist between neighboring PBI's of approximately 20°. In addition, Ghosh *et al.*[22] have shown that the H- and J-forms can be controlled through the selection of peripheral alkyl side-groups. Linear alkyl side-chains favored H-aggregate formation while branched alkyl side chains favored J-aggregate formation. Rotational offsets in both forms were supported by CD measurements, and were required to enable efficient H-bonding and close π–π contact. Based on steric considerations, the H-form was hypothesized to consist of a (0,0) (no slip) helical π-stack, while the J-form was believed to also be helical but with a significant longitudinal slip.

Our analysis is semi-quantitative as it ignores extended Coulombic interactions, electron-hole separations beyond nearest neighbors, the small but finite transition dipole moment of CT states, and the presence of intermolecular, excimer-forming vibrational modes. Although such effects should not significantly alter the essential impact of Coulomb/CT-mediated interferences, they should lead to better agreement with experiment, at the level of what was obtained in our TAT analysis.[41] In Ref. 41 we showed that in the presence of extended Coulombic interactions there remains a dramatic reduction in the exciton bandwidth in HJ aggregates, even though an entirely flat band is likely not possible due to uncompensated Coulomb interactions between non-nearest neighbors. In a future paper we will deal specifically with the effects of states with extended electron-hole separations.

Although we have focused primarily on the *destructive* interference between the CT-mediated and Coulombic couplings, the *constructive* interference operative in HH and JJ aggregates is far more interesting for device applications, since an enhanced exciton bandwidth generally leads to greater exciton mobility and therefore more efficient photovoltaic devices, as one example. The dramatic mobility increase associated with a sub-Angstrom slip predicted in TAT nanopillars[42] is another example of a "J- to H"-aggregate transformation but differs from the mechanism at play in the PBI examples discussed so far. Although, the observed "J-form" remains a two-band HJ aggregate in TAT nanopillars, the predicted "H-form" is quite different than the conventional Coulomb-coupled H-aggregate proposed for the PBI examples. In TAT, the single-band H-form is a HH-aggregate with an exciton bandwidth enhanced by *both* Coulombic coupling and Frenkel/CT mixing. The HJ to HH transformation is associated with a sign change in t_h, making $|t_e - t_h|$ large instead of individually small values of $|t_e|$ and $|t_h|$ as we have suggested for the H-stacks of PBI in Refs. 49, 50, and 40. Designing such HH-aggregates should be possible by inducing small packing modifications (i.e., slips) via chemical tuning[7,20,73,74] or through covalent linking to a scaffold using variable-length spacer groups.[69,75,76] Such high-mobility aggregates with large exciton bandwidths should be readily distinguished from low-mobility aggregates (which are near the null point) by the vibronic ratio in Eq. (17). The larger the ratio deviates from the monomer, the greater is the exciton mobility.[42] We believe that the design paradigm championed here should guide the next generation of organic materials for electronic applications.

ACKNOWLEDGMENTS

We would like to thank F. Wurthner for useful discussions and S. Yagai for providing us with the experimental data used to compose Fig. 7. F.C.S. is supported by the National Science Foundation, Grant No. DMR-1203811.

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